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EDUCATION EXPERIENCE

Guangdong University of Technology, Guangdong, China **09/2020 - 06/2024**

Bachelor of Electronic Science and Technology

- **Language:** Mandarin (native), Cantonese (native), English (CET-6: 528) IELTS (6.5 (6))
- **Relevant coursework:** Advanced Mathematics(85), Theory of Probability and Mathematical Statistics(89), Microelectronic Device and Technology(99), Signal & System(85), C Programming, Embedded System Design and Application, etc.
- **Skills:** MS office, Python, MATLAB, CUDA, NumPy, C/C++/C#, Java, Windows, Linux, HTML, Shell Scripting, etc.
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The Hong Kong University of Science and Technology, China **09/2025 -**

Master of Physics

- **Relevant coursework:** Computational Methods in Science, Statistical Machine Learning, Quantum Materials and Tech nologies for Quantum Devices and Sensors, Applications of Artificial Intelligence in Science, Computational Energy M aterials and Electronic Structure Simulations, etc.
- **Skills:** MS office, Python, MATLAB, CUDA, NumPy, C/C++/C#, Java, Windows, Linux, HTML, Shell Scripting, etc.

RESEARCH EXPERIENCE

AI for Physical Sciences Lab - HKUST, ***Research Assistant*** **08/2025-Present**

Contrastive learning and transformer architecture development for orbital-based learning and electronic structure t heory

- **Content:** Developed advanced machine learning approaches to improve the efficiency and accuracy of electronic structure calculations, addressing the traditional accuracy–cost trade-off in quantum chemistry. The project integrated **orbital-based learning methods (MOB-ML & OrbNet)** with **graph transformer architectures**, and applied **contrastive learning** to build general molecular representations. These methods enable property prediction from low-level theory (e.g., semiempirical, Hartree-Fock) to high-level theory (DFT, Coupled-Cluster), supporting applications in catalyst, drug, and material design.
- **Model Development:** Designed unsupervised **contrastive learning** strategies using orbital descriptors to enhance molecular representation quality.Implemented **graph transformer models** for supervised regression of molecular properties, achieving high accuracy with significantly reduced computational cost.
- **Impact:** Delivered a scalable solution that reduces computational cost by orders of magnitude compared to traditional high-level **electronic structure** methods, enabling efficient prediction for **large-scale molecular systems**.

Guangdong Provincial Key Laboratory of Information Photonics Technology, ***Research Assistant*** **06/2024-08/2025**

3D Convolutional Neural Network-Based Spectral Unmixing for SERS Imaging

- **Content:** We have designed a **3D self-encoder algorithm** based on discrete cosine transform with spatio-temporal global attention mechanism which is designed for realizing a component decomposition unmixing algorithm for scanning Raman imaging of 2D materials **MoS2/WSe2**. We have compared traditional machine learning **NMF** non-negative matrix decomposition and K-hype kernel abundance estimation algorithm with traditional and deep learning algorithms to realize the reconstruction of the **Raman** scanning imaging on **SOTA** level.
- **Publish:** Co-authored a manuscript *(targeting IEEE T-IP, under review)*.
- **Achievement:** This is supported by *National Natural Science Foundation Project (500210181) with Dr. Ping Tang*

Guangdong Provincial Key Laboratory of Information Photonics Technology, ***Research Assistant*** **06/2024-08/2025**

Reinforcement Learning-Driven Super-Resolution Imaging (5x Resolution)

- **Content:** Proposed Markov Decision Process (MDP) framework to iteratively optimise system imaging to achieve super-resolution imaging from **MoS2/WSe2** heterojunction at the electron microscope level, reducing imaging noise by 40%+. Developed a custom RL agent (PyTorch) to shape rewards based on spectral signal-to-noise ratio (SNR).
- **Achievement:** A patent for an invention is being drafted as the first author.

Shanghai Zhangjiang Institute of Mathematics, ***Core Member*** **07/2024 - 10/2024**
Diffusion Models for ISPRS Remote Sensing Data Augmentation

- **Content:** Training a state-of-the-art generative model based on stable diffusion on the ISPRS dataset generates synthetic images to improve semantic segmentation accuracy (mIoU +7.0%), comparing the generative stability of stable diffusion models with adversarial neural network **GANs**, integrating of diffusion-generated data into a U-Net pipeline (PyTorch), and explore the prospects of applying the stable diffusion model to different semantic segmentation networks(**Swin-Unet**, **DeepLab v3**, etc).
- Achievement: This work reduced annotation costs by **30%+**.

PROJECT EXPERIENCE

IoT Temperature and Humidity Sensor Based on STM32F1 and OneNet(coursework) 09/2023 - 11/2023

- The hardware adopts STM32F1 main control chip, ESP8266 module and DHT11 sensor.
- The software is based on Uniapp to develop the mobile terminal.
- responsible for embedded code writing and PCB circuit design, through the MQTT protocol to achieve real-time synchronous monitoring of sensor data to OneNet platform.

Matlab-based GPU programming for optoelectronic image acceleration processing(graduation) 01/2024-05/2024

- This project is based on parallel computing, image processing algorithms, and deep learning networks dataset.
- pre-processed the open source MRI dataset, did work of data enhancement and feature value extraction to achieve the image for classification tasks.
- designed experiments to compare the use of GPU programming and CPU programming for classification of big data image dataset to prove the speed superiority of parallel computing programming.

Unimol2 Advanced Engineering: Graph Transformer & Contrastive Learning Integration 08/2025 - Present

- Enhanced UniMol2 molecular representation learning by using **Graph Transformer architecture** and **contrastive learning loss** to improve generalization and representation quality.
- Implemented full pipeline: model modification, training with orbital-based descriptors, and evaluation on benchmark datasets; optimized training using PyTorch and Hugging Face ecosystem.
- Delivered performance improvements with minimal parameter growth, enabling efficient deployment for large-scale molecular simulations.